

Interesting Case

Pattranan Kusontammarat, MD

**Second Year Fellow, Division of Pediatric Infectious Diseases, Department of
Pediatrics**

Faculty of Medicine, Chiang Mai University

Identification Data

- **A Thai Girl 1 year-6-month-old**
- **Admission date: 23/8/68**
- **Underlying disease: none**

Chief complaint

คลำได้ก้อนที่หน้าท้องข้างซ้ายโตขึ้น 7 days PTA

Present Illness

7 days PTA บิดาคำได้ก้อนที่หน้าท้องด้านซ้าย ขนาดไม่ถึง 1 cm ไม่เจ็บ ไม่บวม ไม่มีไข้ ไม่มีแผลที่ผิวหนัง บริเวณก้อน ไม่มีคลำก้อนได้บริเวณอื่น ไม่มีผื่น เล่นได้ ร่าเริงปกติดี

4 days PTA สังเกตก้อนขนาดใหญ่ขึ้นเยอะ ไม่เจ็บ ไม่บวม แดง ไม่มีไข้ เด็กกินได้ ไม่ซึม จึงพามาตรวจที่รพ.

At OPD EID

Physical examinations:

- **Palpable mass sized 2x3 cm, round, firm, smooth surface, movable, not tender mass at left flank with erythematous skin**, no fluctuation, groin node cannot be palpated

Impression: Infected lipoma

Management: Cephalexin 7 days [45 MKDay], นัด F/U pedsx 1 week

Present Illness

1 day PTA บิดาสั่งเกตว่าก่อนมีขนาดใหญ่ขึ้นเล็กน้อย บวมแดงมากขึ้น ไม่มีไข้ ไม่มีแผล ไม่มีหนอง มารดา
กังวลจึงพากลับมารพ. ก่อนนัด รับประทานยาสม่ำเสมอ ไม่มีผื่นหรือค้ำำได้ก่อนบริเวณอื่น

Past history

- Perinatal: Term female NB, GA 38+5 WK/US, BBW 3390 gm , APGAR 8-9-9 , NL
- No underlying disease
- Deny history of recurrent abscess
- No current medications
- No food/drug allergy
- Vaccination: as EPI
- Family history: no genetic disease in family

Physical Examination

- Vital signs: BT 36.2° C PR 109 bpm RR 28 /min SpO2 97% BW 11 Kg
- HEENT: no pale conjunctiva, anicteric sclera, cervical LN cannot be palpated
- Heart: regular, no murmur
- Lungs: clear and equal BS
- Abdomen: soft, not tender, no distension, spleen and liver cannot be palpated
- Ext: no deformity, groin node cannot be palpated
- Neurological: good consciousness

Physical Examination



Skin:

- Palpable mass sized 3x4 cm, warmth, tenderness on palpation, round, firm, smooth surface, movable at left flank
- Erythematous skin overlying
- No fluctuation

Lab investigation

CBC	Result
Hb (g/dl)	13.8
Hct (%)	42.4
WBC (cell/cu.mm.)	19,310
Neutrophil (%)	33.1
Lymphocyte(%)	57.5
Monocyte(%)	4.2
Eosinophil (%)	4.8
ANC	6,410
AEC	926
Platelet(cell/cu.mm.)	517,000
MCV	74.6

Ultrasound left side abdominal wall

Findings(26/8/68):

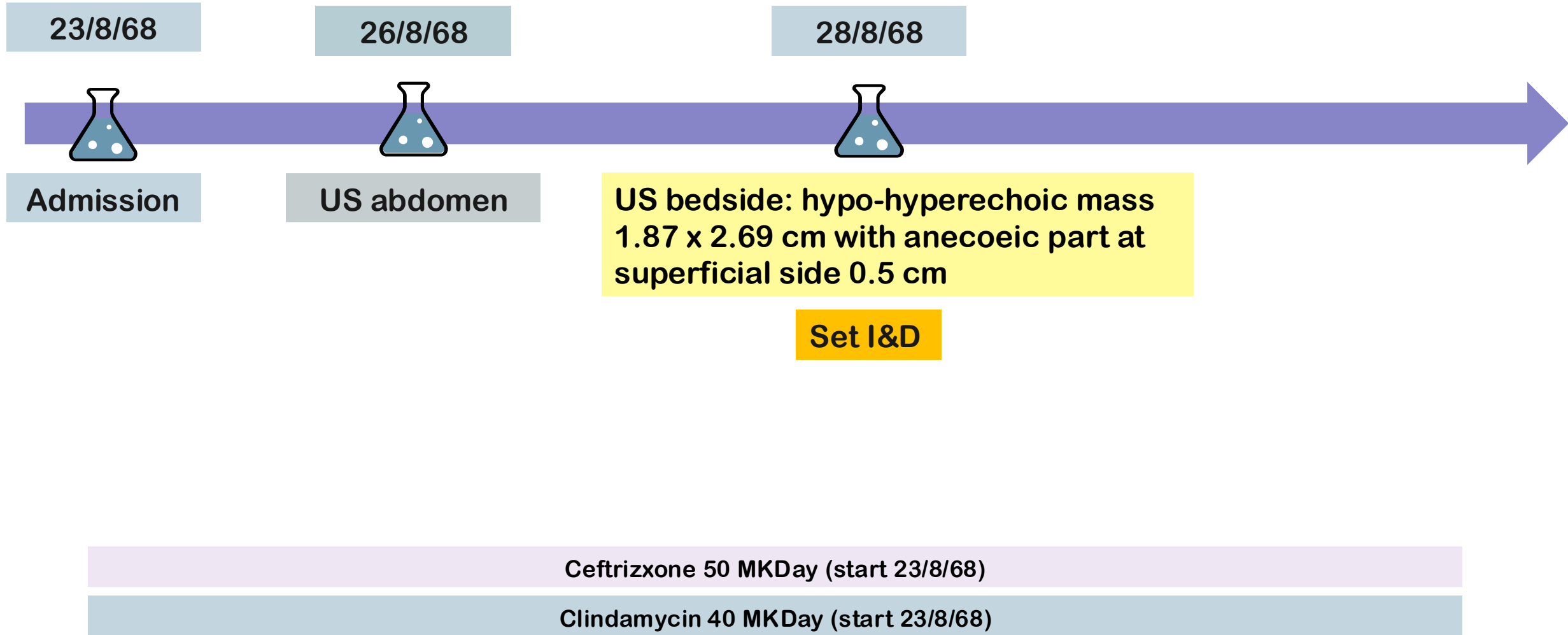
- 2.2x2.8x1.1 cm ill defined heterogenous hypo/hyperechic lesion in the subcutaneous soft tissue at the lower abdominal wall
- No internal vascular flow or calcification
- Increase peripheral vascular flow of this lesion is noted
- Also seen surrounding subcutanouse soft tissue and skin thicken edema

Impression: infected epidermal inclusion cyst in the lower abdominal wall

Management

- Admit and treat **infected epidermal inclusion cyst**
- ABO: ceftriaxone and clindamycin IV
- Plan excision cyst when the infection improved

Timeline



Management

Set I&D bedside 28/8/68: **pus with sebaceous content 1 cc**

- Discharge gram stain: no bacteria
- Discharge culture: no growth

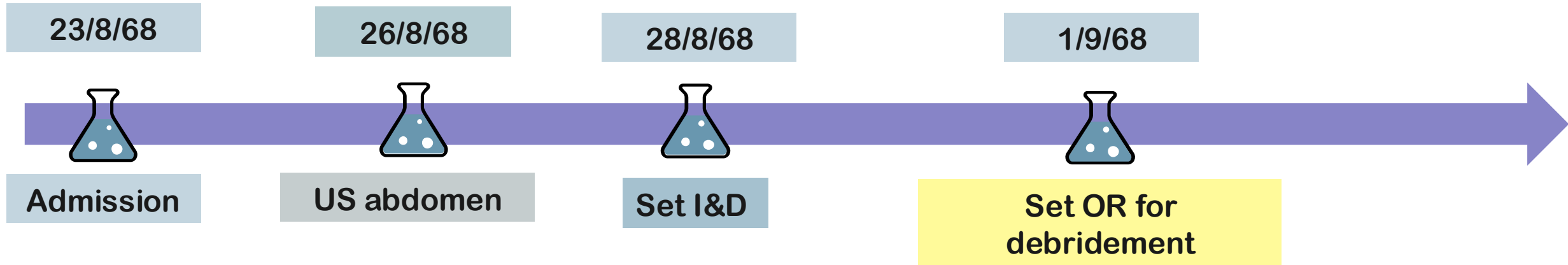


Management

- **Set OR for debridement 1/9/68**



Timeline



Ceftrizxone 50 MKDay (start 23/8/68)

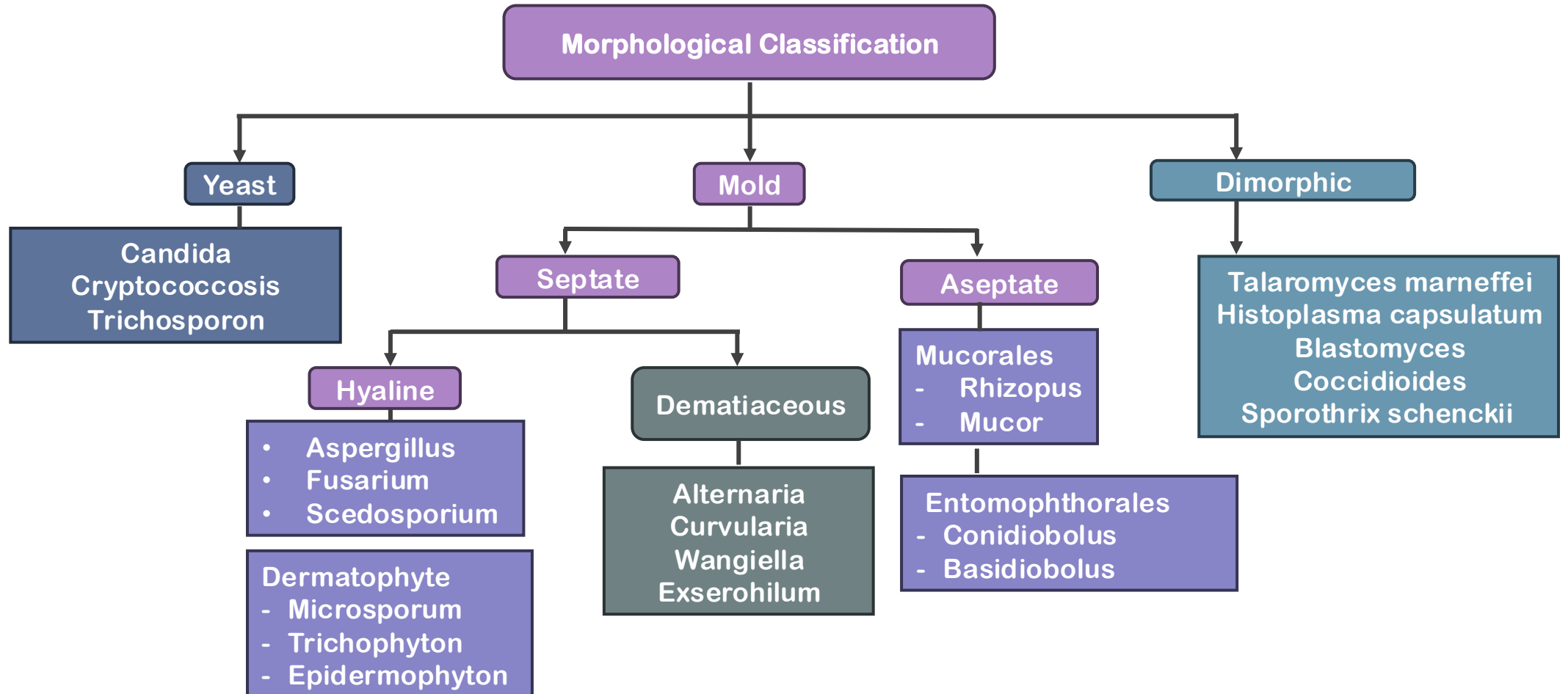
Clindamycin 40 MKDay (start 23/8/68)

Prelim Histopathology Report

Tissue from left flank, debridement:

- **Hyaline septated fungal** infection with tissue invasion
- Differential of fungal type include *Aspergillus* spp. please correlate with tissue culture
- **Non-caseating granulomatous** inflammation with acute necrotizing and chronic inflammation
- Granulation tissue
- Special stain:
 - **GMS & PAS: positive for broad septate hyaline fungal hyphae with acute angle branching within tissue**
 - **AFB: negative for acid fast bacilli**

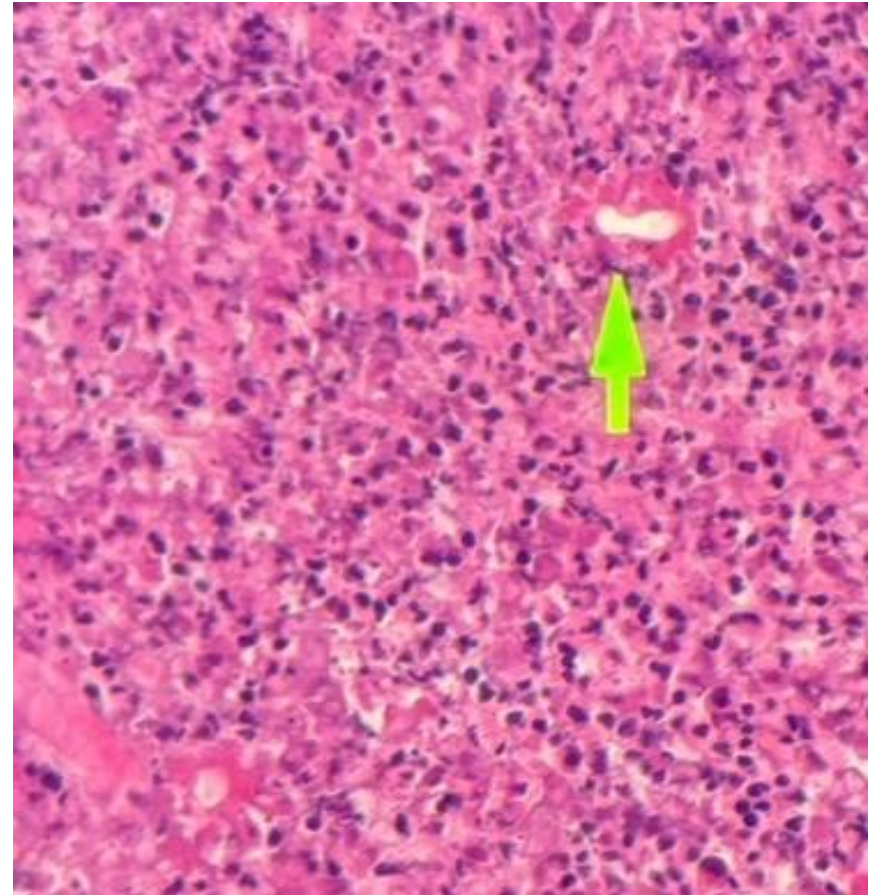
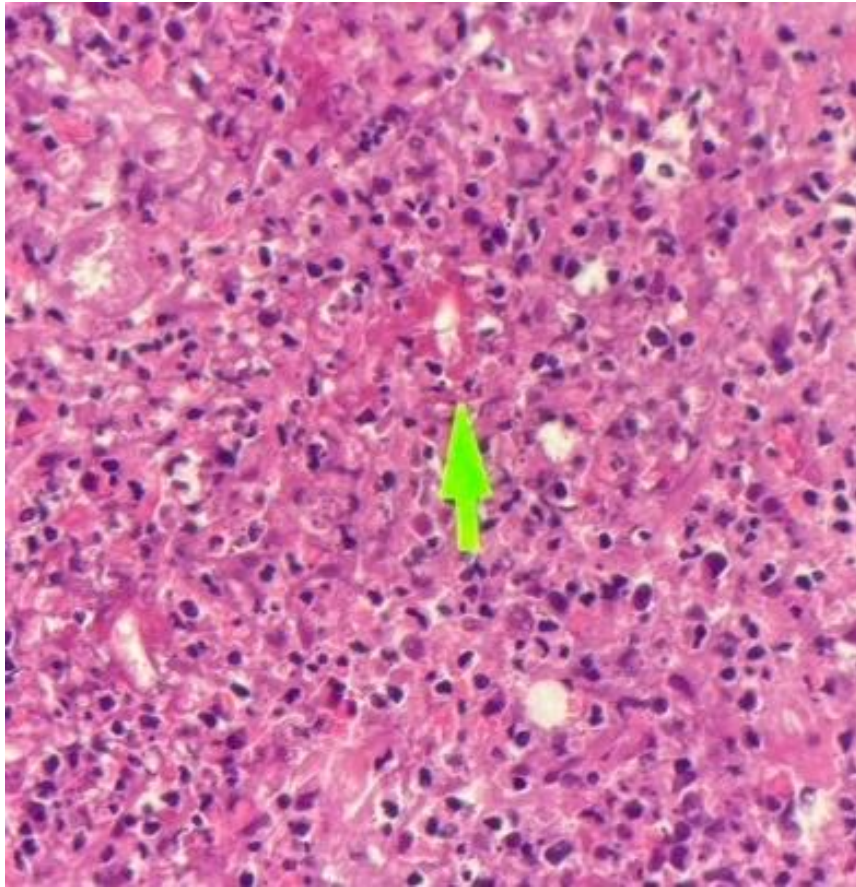
Classification of Fungi



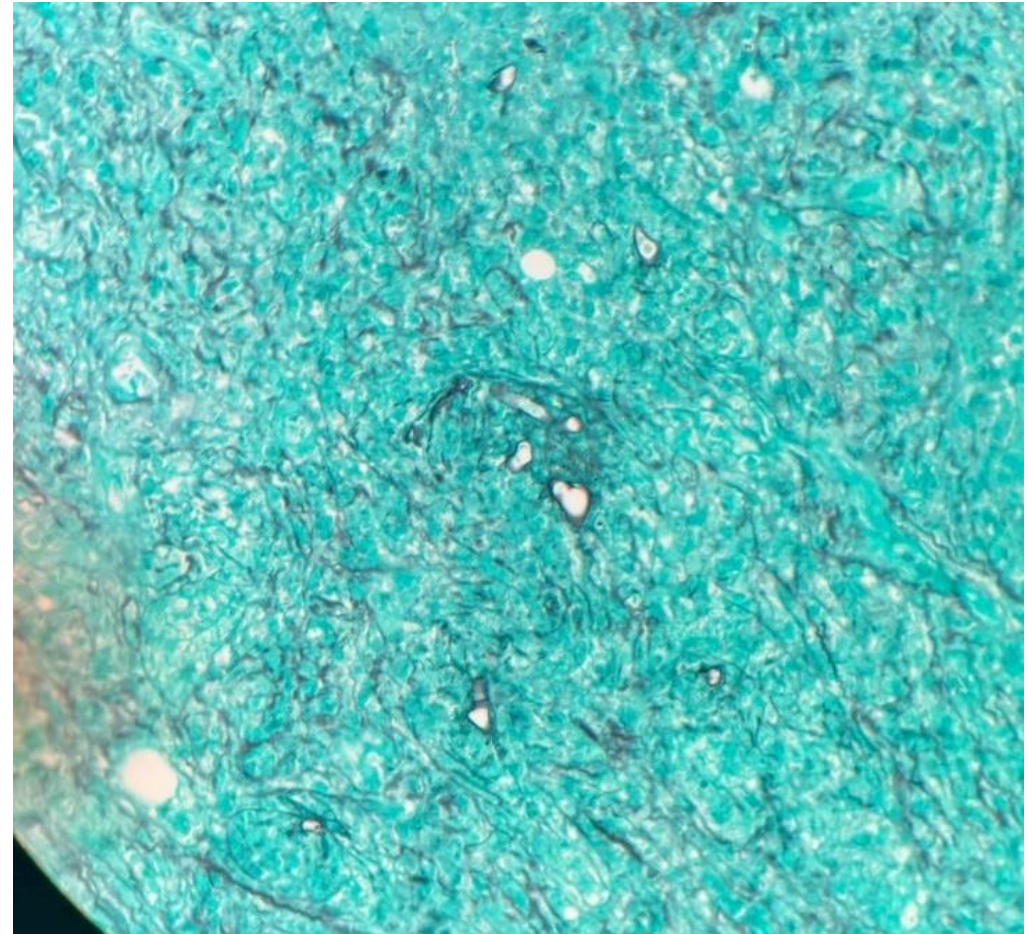
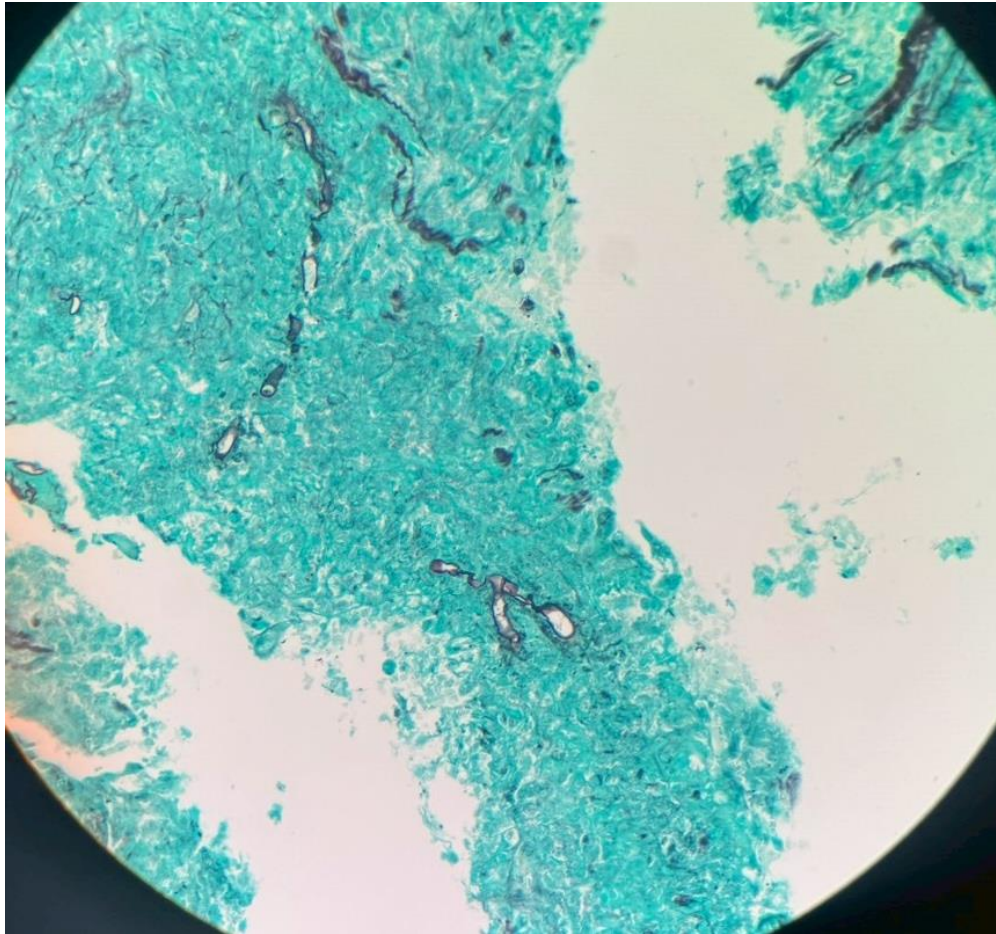
Management

- **Empirical antifungal: amphotericin B 1 MKDay**
- **Suggest tissue for fungal culture and 18srRNA**
- **Review histopathology**

Tissue Histopathology



PAS stain



Tissue Histopathology

(Revised type of fungus after review slides requested by the clinician)

Tissue from left flank, debridement:

- Fungal infection with tissue invasion
- The fungus shows broad hyphae with rare septation and branching surrounded by eosinophilic material, morphologically consistent with *Basidiobolus spp.*

Tissue Investigations

- Wright stain: negative
- Fungus culture: pending
- Tissue 18srRNA: *Basidiobolus spp.*

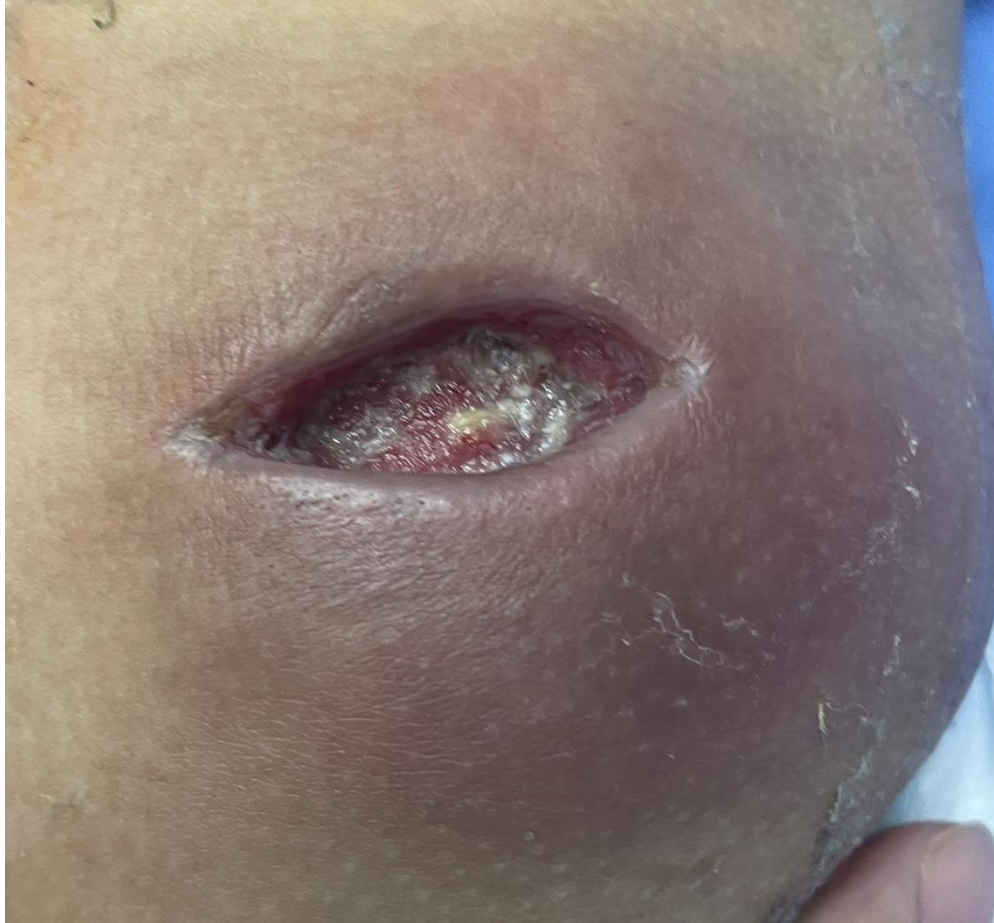
Impression

Subcutaneous Entomophthoromycosis
most likely from *Basidiobolus spp.*

Management

- **Switch to oral itraconazole 5 M/day for at least 3-6 months or until the lesion resolved**
- **Monitor side effects of hepatitis**
- **Continue wound dressing**
- **Consult allergy for work up immune deficiency**
 - **Work up DHR: normal**
 - **Immunoglobulin level: IgA <22.6 (selective IgA deficiency <7 mg/dL)**
 - **Less likely immunodeficiency plan F/U immunoglobulin level next 6 months**

Progression



Basidiobolus spp. Infection

Entomophthoramycosis

- Infections caused by members of the order *Entomophthorales*
- Two genera are pathogenic to humans and they include *Conidiobolus* spp. and *Basidiobolus* spp.
- Entomophthoramycosis is usually, but not always, limited to the skin and subcutaneous tissues in immunocompetent patients
- Cutaneous and subcutaneous entomophthoramycosis in immunocompetent hosts is largely limited to countries within tropical and subtropical climate zones

Clinical manifestations



Chronic rhinofacial
entomophthoramycosis

Conidiobolus spp.



Chronic subcutaneous
entomophthoramycosis



Gastrointestinal
basidiobolomycosis

Basidiobolus spp.

Basidiobolus spp.

- Most human infections are caused by *B. ranarum* and are reported from tropical regions
- *Basidiobolus ranarum* was first described by Eidam in 1886 from frog excrement
- Since then, *Basidiobolus microspores*, *B. lacerate*, *B. haptosporus* (1947) and *B. meristosporus* (1956) have all been described as part of the *B. ranarum* species

Mode of Transmission

- *Basidiobolus* may enter human hosts **via insect bites** or via **minor trauma and subsequent inoculation**
- Mites and other insects carry these organisms on their surface and may directly transfer the fungi to humans during feeding
- Alternatively, these infected insects can be eaten by amphibians which would then excrete the fungus in feces and potentially contaminate even trivial wounds
- In addition, infection may occur from **contaminated soil and vegetation**

Clinical Manifestations

- It commonly affects children
- Presents as typically painless indurated swellings with secondary skin changes, mimicking chronic granulomatous diseases like tuberculosis or even malignancy
- *Basidiobolus ranarum* causes infection of the perineum, thighs, buttocks, thorax, or back and can readily spread from one location to the other
- This infection may also extend subcutaneously across facial planes to involve the arms, thorax and neck
- The classic presentation around buttocks and thighs, basidiobolomycosis is said to have a “bathing suit” distribution
- Deep visceral varieties of entomophthoromycosis have also been documented to be caused by *Basidiobolus* involving the gastrointestinal tract with spread to adjacent areas

Clinical Manifestations

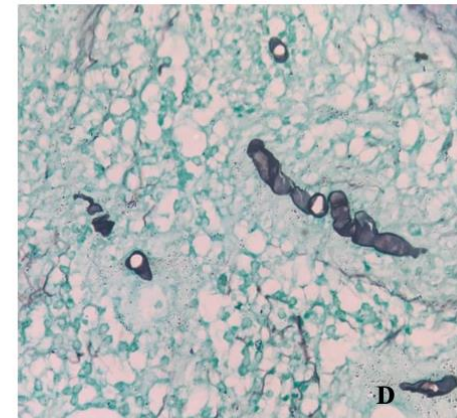
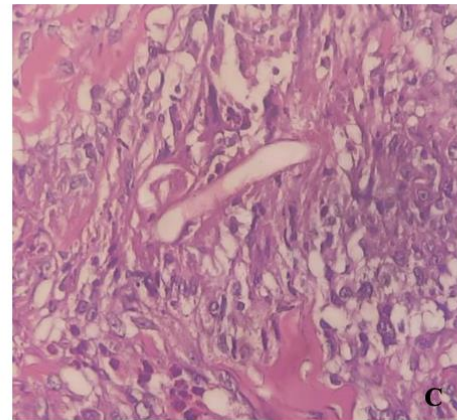
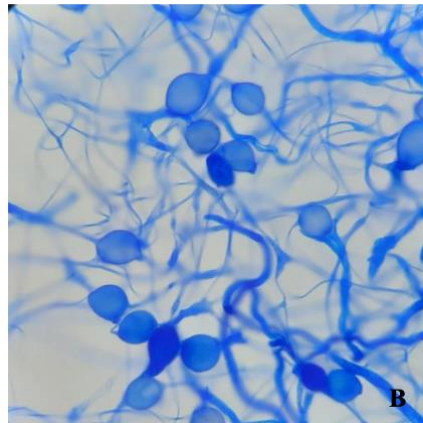
- Although **entomophthoramycosis is typically confined to immunocompetent individuals**, cases have been documented in which immunocompromised patients present with disease, in which cases the organisms tend to behave in a more invasive fashion
- The nodules of basidiobolomycosis **cause erythema and warmth of the overlying skin** due to the underlying inflammation
- The subcutaneous lesions are typically attached to the overlying skin but not the underlying tissues allowing them to remain freely mobile
- *Basidiobolus* can obstruct the underlying lymphatics leading to **lymphedema and enlargement of the affected structure**

Clinical Manifestations

- On examination, the nodules are firm with well-defined borders and tenderness may or may not be present
- Secondary changes in the form of pigmentation, fluctuation with abscess formation, chronic ulceration or scaling are known
- The mass lesions also may extend deeply into muscle and fascia with the potential for retroperitoneal invasion
- Beginning as a subcutaneous nodule, the lesion may expand over months to form lobular subcutaneous masses that can eventually ulcerate the overlying skin
- Without treatment, basidiobolomycosis is usually slowly progressive; however, spontaneous healing is reported

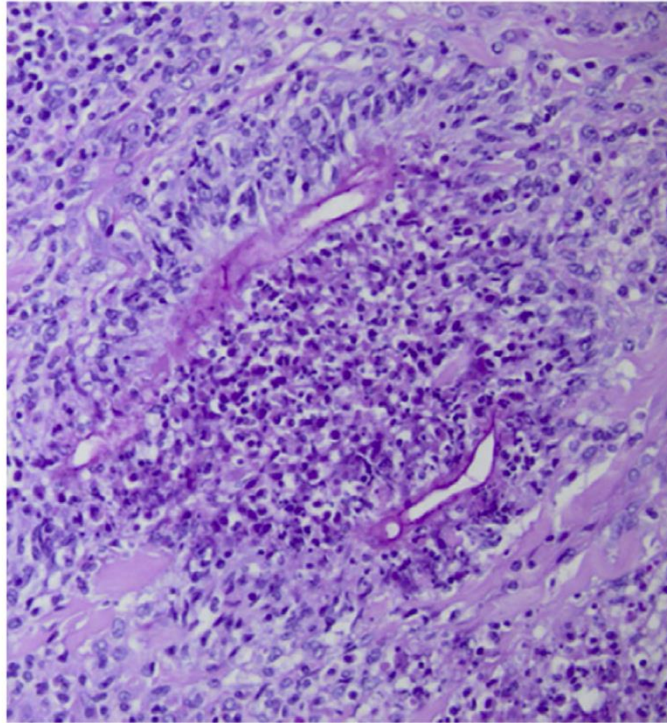
Diagnosis

- Diagnosis of *Basidiobolus* infection is typically by **microbiological culture** with **associated typical histopathological features**
- Our wet mount preparation with **lactophenol cotton-blue stain** found the characteristic **beaked-spores**

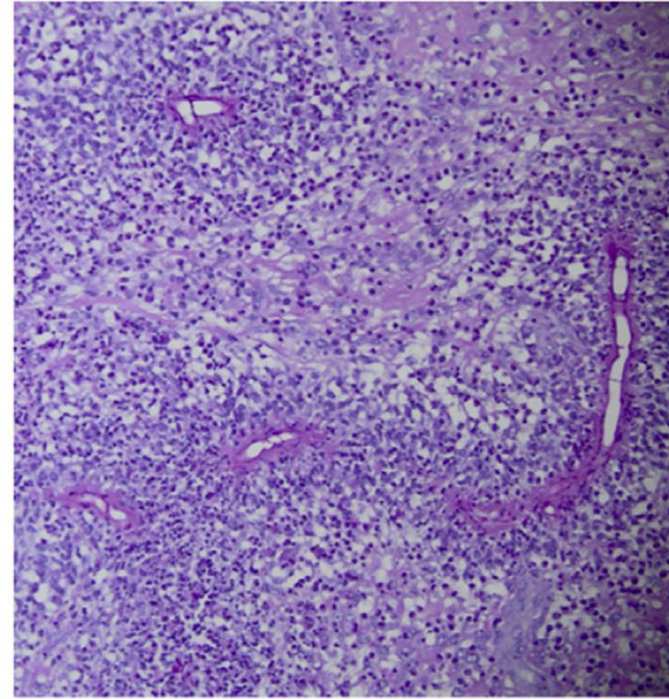


Diagnosis

A



B



A and B) Histopathology with H&E stain is demonstrating the **Splendore-Hoeppli phenomenon** which is describing the intense eosinophilic inflammatory surrounding the ribbon-like hyphae which are sparsely septated

Treatment

- Many cases of the disease **resolve spontaneously**
- Surgical excision is also curative
- The usual treatment of choice has been **potassium iodide (KI)**, the dose being 30 mg/kg daily given as a single daily dose or divided into three daily doses
- **Treatment is continued for 6-12 months**
- Some patients do not respond to KI treatment
- Successful use of **cotrimoxazole** (trimethoprim and sulphamethoxazole) has been reported in several cases

Treatment

- In 1999, Guarro and colleagues showed in-vitro susceptibility of *Basidiobolus* spp. to amphotericin B, fluconazole, flucytosine, itraconazole, ketoconazole and miconazole by a broth microdilution method
- All the *Basidiobolus* isolates demonstrated low MICs to the azole agents which were in contrast to the *Conidiobolus* in the same study, which showed high MICs against all antifungals tested
- Previous case reports describe resolution through treatment with antifungals and without

Antifungal susceptibility of *Basidiobolus* species (9 isolates) and *Conidiobolus* species (8 isolates) [16].

Geometric mean MICs (mg/L) (range)	Amphotericin B	Fluconazole	Flucytosine	Itraconazole	Ketoconazole	Miconazole
<i>Basidiobolus</i> spp	2.7 (0.5–16)	14.8 (2–128)	165.9 (64–256)	1.8 (0.25–32)	1.0 (0.25–4)	3.9 (0.5–8)
<i>Conidiobolus</i> spp	3.1 (0.5–8)	107.5 (32–128)	234.6 (128–256)	11.3 (0.25–32)	20.7 (4–32)	11.3 (4–16)

Case Reports

A 15-year-old Thai woman from Pathumthani

Chief complaint: Indurated slow-growing subcutaneous plaque on left popliteal fossa for 2 months



Present illness: Two months ago, patient developed small erythematous **subcutaneous nodule** on left popliteal fossa. The lesion **gradually increased in size** and formed **painless indurated plaque** without ulceration. There was no history of trauma or insect bite. She went to a private hospital which **incision and drainage** was performed. Tissue was also sent for histopathology. **Intravenous antibiotics were prescribed for 4 days but the lesion did not improve**

Past history: Healthy

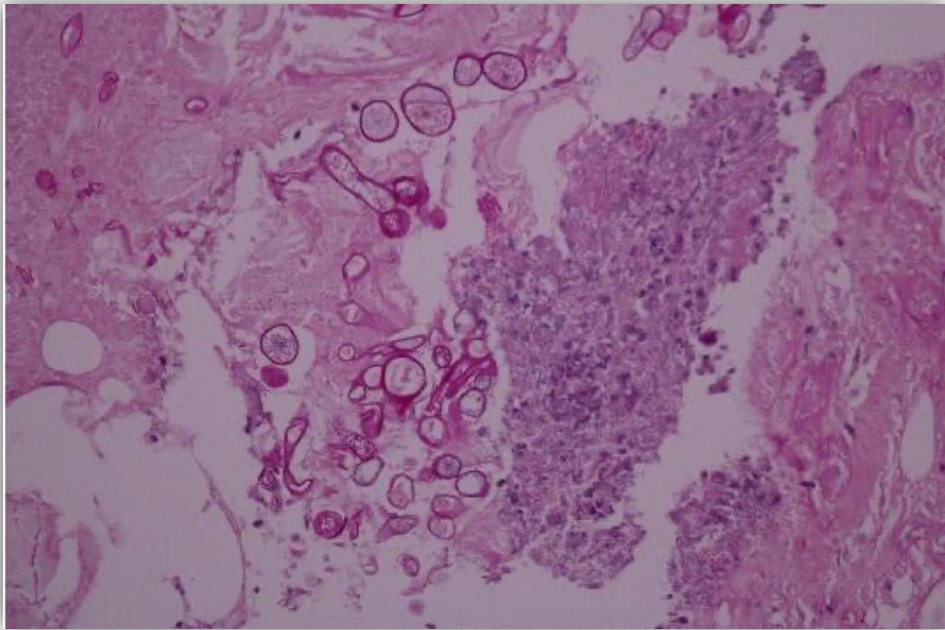
Physical examination:

Skin : Solitary erythematous to brownish indurated plaque with central ulcer, sized 1x2 cm in diameter

Case Reports

A 15-year-old Thai woman from Pathumthani

Chief complaint: Indurated slow-growing subcutaneous plaque on left popliteal fossa for 2 months



Histopathology: diffuse infiltrate of lymphocytes, histiocytes, admixed with some **multinucleated giant cells, neutrophils, eosinophils in deep dermis and subcutaneous tissue**

- fat necrosis, some with basophilic amorphous material giving the feature of saponification in subcutaneous tissue
- **Large thin wall broad hyphae with multinucleated histiocyte and necrotic foci**

GMS&PAS stains: **thin wall broad hyphae within subcutaneous tissue**

Investigations: CBC: WBC 8,170/uL (PMN 61%, L29%, Mo 5%, Eo 5%), Hb 11.6g/dL, Hct 37.1%, Platelet 475,000/uL

Tissue culture for fungus: *Basidiobolus spp.*

Diagnosis: Basidiobolomycosis

Treatment: Itraconazole 200 mg oral twice daily for 3 months

Case Reports

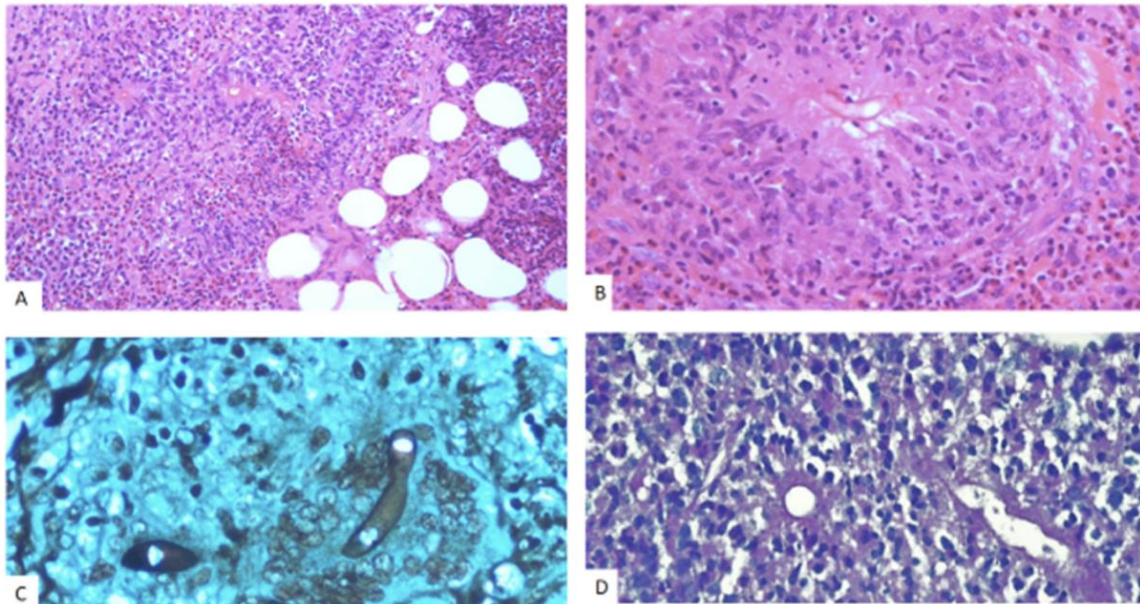
A five-year old girl presented with a history of swelling over the right lower back for three months. Initially the swelling was peanut sized and progressively increased over three months



- It was **diagnosed to be an abscess**, and a drainage was attempted but abandoned due to a bloody aspirate
- On examination:
 - **10×8 cm firm, indurated, fixed scaly painless lesion was noted in right lower back** just above the iliac crest
 - The surface was erythematous without any rise in local temperature
 - Posteromedial aspect of right upper thigh had another 3 × 3 cm lesion

Case Reports

A five-year old girl presented with a history of swelling over the right lower back for three months. Initially the swelling was peanut sized and progressively increased over three months



- Microscopy(KOH): **Broad, hyaline, pauci-septate hyphae**
- Culture: Waxy colonies, cerebriform center, no aerial mycelium; LPCB: beaked zygosporangia
- Histopathology: Suppurative granulomata with eosinophils, **Splendore-Hoeppli phenomenon, broad aseptate hyphae**
- Molecular analysis: *Basidiobolus meristosporus*
- Diagnosis: *Basidiobolus ranarum*
- Treatment: **Oral Saturated Solution of Potassium Iodide (SSKI)**

Case Reports

- The child was initiated on SSKI, 2 drops twice daily (200mg/day in 4 drops). The plan was to escalate sequentially to 3 drops thrice daily (150mg/dose) to reach the recommended dose (450mg/day)



- On reaching 300mg/day dosage, she developed symptomatic hyperkalemia (serum potassium of 5.10 mEq/L) with tall T waves on EKG
- SSKI was stopped and oral Itraconazole (5mg/kg/day) was started which was well tolerated with good clinical response
- Response was dramatic with almost 70% resolution of the lesion within three weeks of therapy
- By the end of six weeks there was near complete clinical resolution with minimal residual lesion on ultrasonography

Summary of published case reports and series of *Basidiobolus spp* infections, clinical presentation, treatment and outcome

- The literature describes 3 fatal cases of *Basidiobolus* infections amongst the 19 cases
- Two of the patients had progressive disease without antifungal therapy, although the cause of death was not directly attributable to the *Basidiobolus spp.* infection
- The third patient had a severe obstructive bowel lesion as well as a hepatic lesion and died of septic shock despite commencing broad-spectrum antifungal therapy prior to deterioration

Summary of published case reports and series of *Basidiobolus spp* infections, clinical presentation, treatment and outcome.

Reference	Year	Country	Age (years)/ Sex	Site of Infection	Surgery	Antifungal therapy/ route/ duration	Outcome
van den Berk GEL	2006	The Netherlands	61 M	Gastrointestinal	Colonoscopy, hemicolectomy	Amphotericin B	deceased
Singh R	2008	India	11 F	Nasal cavity with extension into ethmoid, maxillary sinuses and extradural extension	Nasal and sinus endoscopy with biopsy	Unknown	Lost to follow-up
Anand M	2010	India	3 M	Skin- thigh	Excision	Potassium iodide/oral/1 week then Itraconazole/oral/6 months	Complete resolution
Ezzedien Rabi M	2011	South Arabia	25 F	Gastrointestinal, intramural colonic mass	Subtotal colectomy, jejunal loop resection, ileosigmoid colostomy	Itraconazole/ oral/ unknown duration	Complete resolution
Al Jarie A.	2011	Saudi Arabia	6 M	Gastrointestinal, intrahepatic	Biopsy only	Amphotericin B then Itraconazole/oral/ unknown	Complete resolution
			13 F	Gastrointestinal Recurrence of intrabdominal mass	Laparotomy and mass resection Hemicolectomy and ileocolic anastomosis for recurrence	Initially declined antifungal therapy Recurrence treated with Amphotericin B then itraconazole/oral/ 12 months	Complete resolution
			8 F	Gastrointestinal	Biopsy and cholecystojejunostomy	Amphotericin B then Itraconazole/ oral/ 18 months	Complete resolution
			8 M	Nasopharyngeal airway including maxillary and ethmoid sinuses	Biopsy Tracheostomy due to upper airway reconstruction	Amphotericin B then Itraconazole/ oral/ 12 months	Complete resolution
			10 M	Pulmonary	Bronchoscopy and biopsy	Amphotericin B then Voriconazole/ unknown/ unknown	Complete resolution
Gordon CL	2012	Australia	53 M	Right leg with inguinal lymphadenopathy	Skin biopsy, lymph node excision biopsy	Itraconazole/ oral/ 12 months	Complete resolution
			5 M	Right chest wall skin lesion	Skin biopsy	Terbinafine/ oral/ 12 months Fluconazole/ oral/ unknown duration	Lost to follow-up
Jayanth ST	2013	India	58 F	Skin soft tissue, right gluteal region	Excision	Potassium iodide/ oral/ 6 months	Complete resolution
Rose SR	2013	United States of America	67 M	Gastrointestinal	Biopsy, debulking	Fluconazole (unknown duration) Posaconazole/ oral/ 12 months	Complete resolution
Zabolinejad N	2014	Iran	3 M	Gastrointestinal	Laparotomy and resection	Posaconazole/ oral/ 3 months; then Itraconazole/ oral/ 12 months	Complete resolution
Albaradi BA	2014	Saudi Arabia	11 M	Gastrointestinal	Biopsy only	Voriconazole/ oral/ 12 months	Complete resolution
Saka B	2017	Togo	5 M	Skin over buttock, back and right flank	Biopsy	Ketoconazole/ oral/ 8 weeks	Complete resolution
Brun LVC	2018	Benin	3 F	Skin- right flank	Excision	None known	Complete resolution
			43 M	Skin- lower limb	Excision	None known	Deceased unknown cause
			3 M	Skin- lower limb	Biopsy	None known	deceased

Take Home Message

- *Basidiobolus* spp. is a rare cause of **subcutaneous and gastrointestinal entomophthoramycosis**, usually affecting **immunocompetent hosts**
- Diagnosis requires a combination of microscopy, culture, histopathology, and molecular tools, as routine methods may be inconclusive
- Key histopathological finding is **Splendore-Hoeppli phenomenon with eosinophilic granulomatous inflammation**
- Early and accurate diagnosis helps to avoid unnecessary expensive pharmacotherapy and mutilating surgeries



Thank You